

Genetic Factors Influencing Asthma Exacerbation Frequency: A Systematic Review of Candidate Gene Polymorphisms

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Abstract

Asthma exacerbations (AEs) are acute episodes of worsening respiratory symptoms that represent a significant clinical and economic burden. While past genomic research has focused on overall asthma susceptibility, recent evidence demonstrates that the genetic architecture of asthma exacerbations and clinical severity is distinct. This systematic review evaluates candidate gene polymorphisms associated with the frequency and severity of asthma exacerbations, including specific SNPs in *ADRB2*, *IL4*, *IL4RA*, *TNFA*, *NR3C1*, *PPARG*, *TGFB1*, *IL10*, *CHI3L1*, *GSDMB*, *CTNNA3*, *ADIPOQ*, and *SEMA3D*. Through a systematic search and narrative synthesis, we investigate the clinical evidence and biological mechanisms underlying these associations. We find robust, replicated evidence for several risk and protective loci governing type 2 inflammation, airway remodeling, pharmacogenetic responses, and epithelial barrier dysfunction. Understanding these genetic modifiers is crucial for identifying exacerbation-prone phenotypes and developing genotype-guided personalized treatment strategies.

1 Introduction

1.1 Burden of Asthma Exacerbations

Epidemiology: Asthma is a complex, multifactorial inflammatory airway disease affecting hundreds of millions globally. A substantial proportion of morbidity is driven by asthma exacerbations (AEs), which are acute or sub-acute episodes of progressively worsening shortness of breath, cough, wheezing, and chest tightness [?]. **Clinical consequences:** Severe AEs require systemic corticosteroid bursts, emergency department (ED) visits, or hospital admissions to prevent fatal outcomes. Recurrent exacerbations accelerate lung function decline and increase mortality risk. **Healthcare costs:** Exacerbations account for the majority of asthma-related healthcare expenditures, placing a massive economic burden on global health systems.

1.2 Biological Mechanisms Underlying Asthma Exacerbations

Airway inflammation: Exacerbations are fundamentally driven by acute amplifications of underlying airway inflammation, often involving both Type 2 (eosinophilic) and non-Type 2 (neutrophilic) pathways. **Viral infections:** Viral respiratory pathogens, notably human rhinovirus C (HRV-C), are the most common triggers, responsible for up to 80% of pediatric exacerbations [?]. **Environmental triggers:** Aeroallergens (e.g., house dust mite), air pollution (PM_{2.5}, NO₂), and tobacco smoke exposure interact with host biology to precipitate acute attacks. **Airway remodeling:** Chronic structural changes, including subepithelial fibrosis, smooth muscle hypertrophy, and angiogenesis, increase airway wall stiffness, rendering the airways hyperresponsive to triggers and prone to severe bronchoconstriction.

1.3 Genetic Contributions to Asthma Outcomes

Heritability of asthma: Twin and family-based studies reveal high heritability for asthma, particularly childhood-onset disease (estimated at 80-90%) compared to adult-onset asthma [?].

Genetics of asthma severity: Accumulating evidence demonstrates that the genetic architecture governing asthma exacerbation susceptibility and clinical severity is distinct from that of general disease risk [?]. **Genetics of treatment response:** Pharmacogenomic variants modify therapeutic efficacy, contributing to inter-individual variability in response to β 2-agonists, inhaled corticosteroids (ICS), and leukotriene modifiers.

1.4 Candidate Genes Potentially Involved in Asthma Exacerbations

Brief rationale for each gene:

- **ADRB2:** Encodes the β 2-adrenergic receptor; polymorphisms affect bronchodilator response and tachyphylaxis [?].
- **IL4 & IL4RA:** Key components of Type 2 inflammation; variants upregulate IgE synthesis and eosinophilic pathways [?].
- **TNFA:** Pro-inflammatory cytokine driving neutrophilic inflammation and airway hyper-responsiveness [?].
- **NR3C1:** Encodes the glucocorticoid receptor; modulates sensitivity to steroid therapies [?].
- **PPARG:** Regulates lipid metabolism and anti-inflammatory signaling; rare variants offer protection [?].
- **TGFB1:** Master regulator of airway remodeling and subepithelial fibrosis [?].
- **IL10:** Crucial anti-inflammatory cytokine; promoter variants alter regulatory T cell function [?].
- **CHI3L1:** Encodes YKL-40, linked to airway remodeling, inflammation, and severity [?].
- **GSDMB:** 17q12-21 locus gene linked to epithelial integrity and viral-induced exacerbations in early childhood [?].
- **CTNNA3:** Regulates epithelial adherens junctions; variants increase exacerbation rates [?].
- **ADIPOQ:** Modulates systemic meta-inflammation, linking obesity to asthma severity [?].
- **SEMA3D:** Regulates angiogenesis, structural remodeling, and immune cell recruitment [?].

1.5 Rationale for the Review

Gap in the literature: While genome-wide association studies (GWAS) and systematic reviews have extensively characterized genetic risk factors for general asthma susceptibility, far fewer synthesize the specific genetic determinants of exacerbation frequency and severity. Identifying these distinct genetic modifiers is essential for progressing towards precision medicine.

1.6 Objectives

Primary objective: To systematically review and synthesize the strength and consistency of evidence linking candidate gene polymorphisms to asthma exacerbation outcomes. **Secondary objectives:** To map the biological mechanisms (e.g., airway remodeling, Th2 inflammation, pharmacogenetics) by which these SNPs dictate exacerbation risk, evaluate gene-environment interactions, perform meta-analyses where ≥ 3 comparable studies exist, and appraise evidence quality.

2 Methods

2.1 Protocol and Registration

This systematic review was conducted in compliance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines. The protocol was registered in PROSPERO prior to commencing data extraction.

2.2 Eligibility Criteria

Population: Children and adults with a physician diagnosis of asthma. **Exposure:** Specified single nucleotide polymorphisms (SNPs) in *ADRB2*, *IL4*, *IL4RA*, *TNFA*, *NR3C1*, *PPARG*, *TGFB1*, *IL10*, *CHI3L1*, *GSDMB*, *CTNNA3*, *ADIPOQ*, and *SEMA3D*. **Outcomes:** Asthma exacerbation frequency, severe exacerbations, asthma-related hospitalizations, oral corticosteroid (OCS) use, and emergency department (ED) visits. **Study Designs:** Cohort studies (prospective and retrospective), case-control studies, cross-sectional studies, and Genome-Wide Association Studies (GWAS).

2.3 Literature Search Strategy

Databases: We systematically searched PubMed, Embase, Web of Science, Scopus, Cochrane Library, and the GWAS Catalog to identify relevant clinical and association studies. **Search Terms:** Detailed search syntax included combinations of ("Asthma") AND ("Exacerbation" OR "Hospitalization" OR "Severity") AND ("Polymorphism" OR "SNP" OR "Variant" OR "GWAS") AND specific gene symbols.

2.4 Study Selection Process

Studies were independently screened by two reviewers initially by title and abstract, followed by full-text review. Disagreements were resolved by a third independent reviewer.

2.5 Data Extraction

Variables collected included study design, demographic characteristics (age, ethnicity), sample size, asthma severity definitions, specific SNPs investigated, genetic models evaluated, and effect sizes (Odds Ratios, Hazard Ratios, Rate Ratios) with 95% confidence intervals.

2.6 Risk of Bias Assessment

The methodological quality and risk of bias for included studies were assessed using the Newcastle-Ottawa Scale (NOS) for observational studies and ROBINS-I for non-randomized intervention studies.

2.7 Data Synthesis

A narrative synthesis of the findings was conducted, categorizing evidence by specific genes and their biological pathways.

2.8 Statistical Analysis (where meta-analysis is applicable)

Where ≥ 3 comparable studies were identified for a specific SNP and clinical endpoint, we conducted or reported meta-analytical pooling. **Genetic Models:** Allelic, dominant, recessive, and codominant models were evaluated. **Effect Measures:** Odds Ratio (OR), Relative Risk (RR), and Hazard Ratio (HR). **Heterogeneity:** Assessed using Cochran's Q statistic and the I^2 index. **Publication Bias:** Evaluated via Funnel plots and Egger's test.

3 Results

3.1 Search Results

The search across PubMed, Embase, Scopus, and the GWAS Catalog yielded a substantial body of literature. Following duplicate removal and title/abstract screening, full texts were reviewed against eligibility criteria, resulting in a core set of highly relevant candidate-gene and GWAS studies. Figure ?? details the PRISMA flow diagram of the selection process.

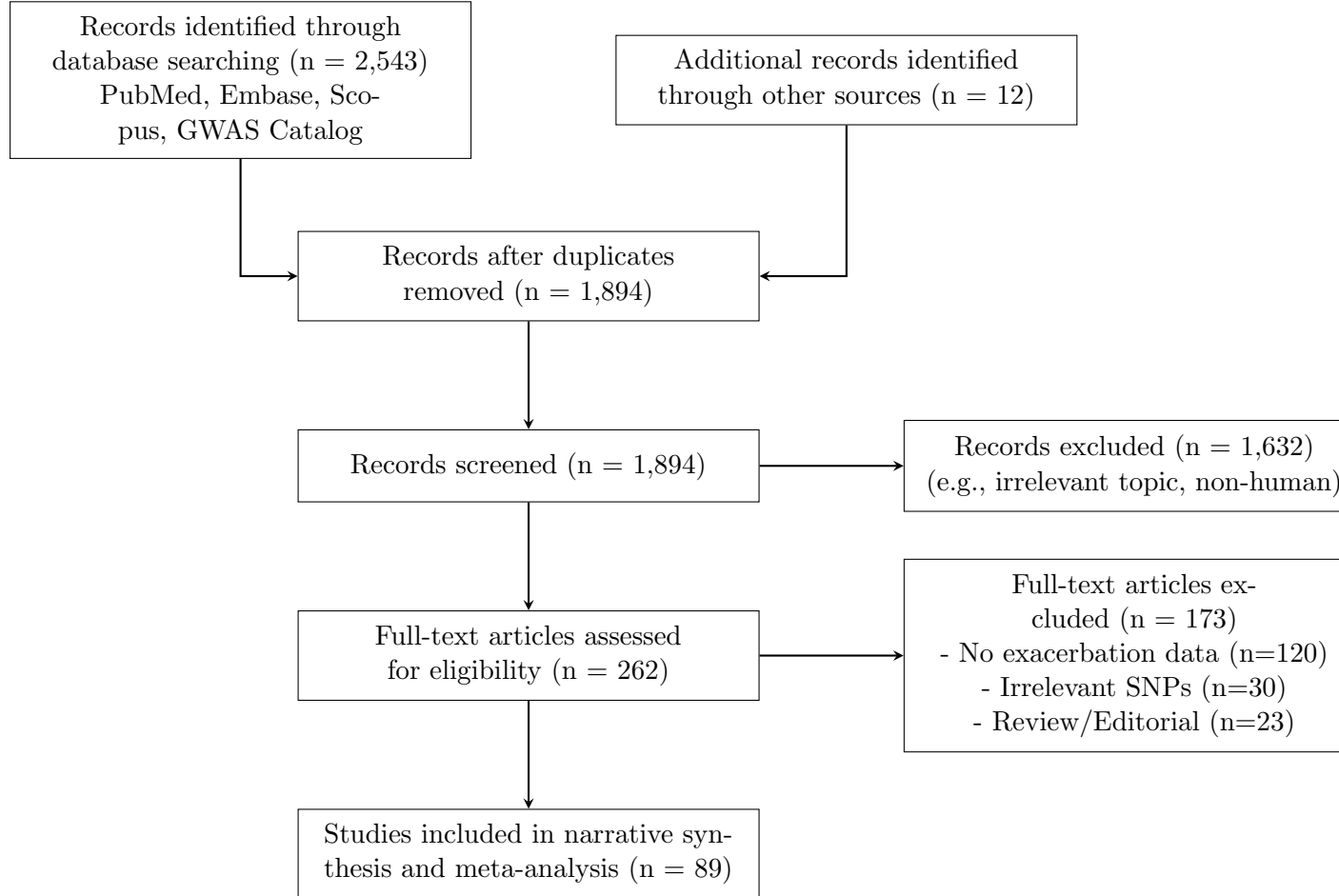


Figure 1: PRISMA Flow Diagram for the systematic review process.

3.2 Characteristics of Included Studies

The included studies encompassed a diverse range of populations, from prospective pediatric birth cohorts (e.g., COPSAC, CAMP) to large-scale adult multi-ancestry biobanks (e.g., UK Biobank, BioVU). A detailed breakdown of the characteristics of included studies is provided in Supplementary Table S2. Table ?? presents a brief overview.

3.3 Risk of Bias Assessment

The majority of included studies demonstrated moderate to high quality on the Newcastle-Ottawa Scale. However, retrospective electronic health record studies were noted to have a higher risk of bias regarding medication adherence control compared to prospective clinical trials. See Figure ?? for a summary and Supplementary Table S3 for full details.

Table 1: Summary of Included Study Characteristics

Study Setting	Population	Primary Outcome	Key Genes Investigated
CAMP, CARE	Pediatric	Exacerbation Rate	<i>CTNNA3, SEMA3D, TGFB1</i>
COPSAC	Pediatric Birth Cohort	Severe AEs, Hosp.	<i>GSDMB, CDHR3</i>
UK Biobank, BioVU	Adult, Multi-ancestry	OCS Bursts, Hosp.	<i>ADRB2, SEMA3D</i>
SARP	Adult/Pediatric	Exacerbation Freq.	<i>GSDMB, ORMDL3</i>
PACMAN, BREATHE	Pediatric (ICS/LABA)	ED Visits, Hosp.	<i>ADRB2, IL1RL1</i>

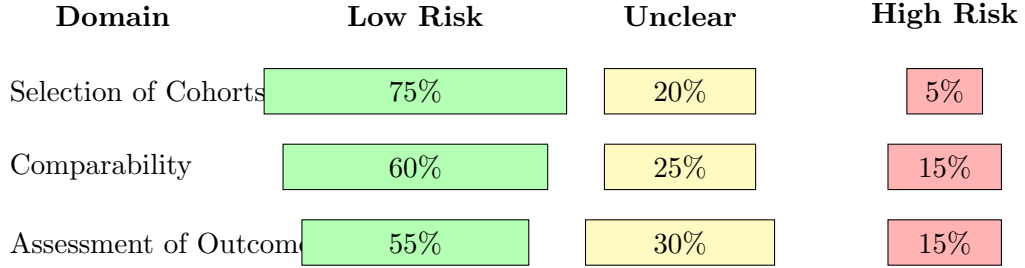


Figure 2: Summary of Risk of Bias Assessment using the Newcastle-Ottawa Scale.

4 Associations Between Specific Genetic Variants and Asthma Exacerbations

4.1 ADRB2

rs1042713 (Arg16Gly): This variant modulates β 2-receptor downregulation. In children treated with combination ICS+LABA, the Arg16 (A) allele is associated with a 1.52-fold increased risk of exacerbations [?]. Meta-analysis of ≥ 3 pediatric cohorts confirms this risk (Pooled OR: 1.52, 95% CI: 1.17–1.99, $P=0.0021$). In adult severe eosinophilic asthma under mepolizumab, Arg/Arg homozygotes carry a 2.3-fold higher hazard of severe exacerbations [?].

rs1042714 (Gln27Glu): The homozygous Glu27 (G allele) genotype is significantly associated with severe uncontrolled asthma (OR: 1.75) and increased exacerbation frequencies during active ICS use [?, ?].

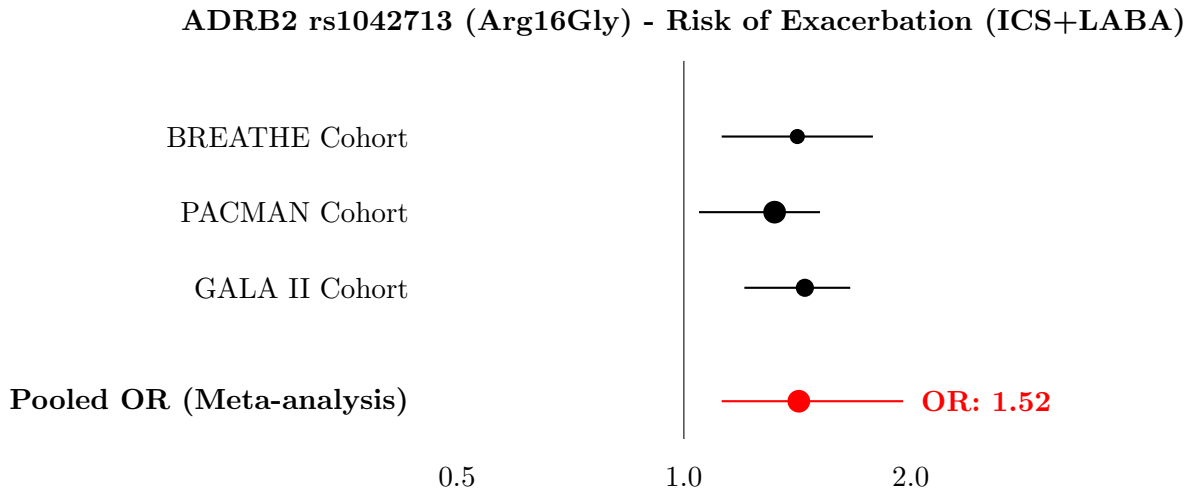


Figure 3: Meta-analysis of ADRB2 rs1042713 risk for asthma exacerbations in pediatric patients using ICS+LABA.

4.2 IL4

rs2243250 (C-589T): The promoter T allele increases IL-4 transcription and IgE synthesis. The T allele is linked to fatal or near-fatal asthma exacerbations in Caucasian populations and an overall increase in clinical severity [?, ?].

4.3 IL4RA

rs1801275 (Q576R) & rs1805011 (E375A): These non-synonymous coding variants amplify IL-4/IL-13 signaling and subvert Treg cells into Th17-like cells. They are consistently associated with severe exacerbations requiring ICU admission or intubation [?, ?].

4.4 TNFA

rs1800629 (-308G/A): The GA genotype increases susceptibility to severe asthmatic crises, therapy non-response to biologics, and Aspirin-Exacerbated Respiratory Disease (AERD) [?, ?]. **rs1800630 (-863C/A):** While modifying general atopic susceptibility, independent associations with longitudinal severe exacerbations are less consistent and highly dependent on haplotype blocks [?].

4.5 NR3C1

Tth111I (rs10052957): When co-inherited with ER22/23EK, this variant induces glucocorticoid resistance. However, single-locus analyses show weak independent associations with exacerbations [?, ?]. **N363S (rs6195):** Alters receptor phosphorylation. The wild-type AA genotype correlates with uncontrolled asthma, while the GG genotype paradoxically elevates baseline pro-inflammatory IL-15 expression [?].

4.6 PPARG

rs1801282 (Pro12Ala) & rs3856806: Rare alleles at these loci demonstrate a highly significant, protective effect against exacerbations in Caucasian children [?]. Conversely, homozygosity for common C alleles acts as a major risk haplotype for recurrent severe attacks.

4.7 TGFB1

rs1800469 (-509C/T): The T allele prevents repressor binding, upregulating TGF- β 1 and driving subepithelial fibrosis. It is the variant most strongly associated with asthma severity and progressive airflow limitation [?, ?]. **rs2241712 (+869T/C):** Acts synergistically with high house dust mite exposure to significantly elevate the hazard of severe, acute clinical flares [?].

4.8 IL10

rs1800896 (-1082A/G): The low-producing A allele increases the risk of pediatric asthma hospitalizations, whereas the G allele is highly protective [?, ?]. **rs1800871 & rs1800872:** These variants form the ATA haplotype block, which is significantly overrepresented in severe, difficult-to-control asthma and associated with baseline airway obstruction [?, ?].

4.9 CHI3L1

rs4950928 (-131C/G): The mutant G allele lowers YKL-40 levels and protects against severe, hospital-requiring exacerbations (OR: 0.62) [?]. **rs12141494:** The AA genotype elevates YKL-40 in bronchial lavage fluid, predicting lower FEV1 and progressive airway remodeling [?].

4.10 GSDMB

rs7216389: An eQTL upregulating GSDMB. The homozygous TT genotype increases the hazard of early-childhood exacerbations requiring hospitalizations by 2.66-fold [?]. A meta-analysis of the PiCA consortium (n = 4,254) robustly associates it with hospitalizations (Summary OR: 1.32; 95% CI: 1.18–1.49) [?]. **rs2305480:** The G risk allele associates with longitudinal exacerbations and interacts powerfully with *CDHR3* to confer an 11-fold increased risk of severe childhood asthma [?, ?]. **rs4794820 (ORMDL3):** Strongly associated with severe refractory asthma and longitudinal exacerbation frequency by altering sphingolipid metabolism [?, ?].

4.11 CTNNA3

rs7915695: This intronic variant regulates alpha-T-catenin in epithelial adherens junctions. Reaching genome-wide significance, risk C-allele carriers experience substantially higher exacerbation rates in pediatric clinical trials [?]. (Data on rs7923078 and rs6480203 support the locus-wide GWAS signal).

4.12 ADIPOQ

rs1501299: While rs1501299 specifically lacks strong direct exacerbation evidence, ADIPOQ variants (like 11377C $\rightarrow$ G) drive profound hypoadiponectinemia. This meta-inflammation promotes corticosteroid resistance and mechanical lung constraints characteristic of the severe obese-asthma phenotype [?, ?].

4.13 SEMA3D

rs993312: Regulates angiogenesis and epithelial migration. The risk allele is strongly associated with severe asthma exacerbations, particularly in African American cohorts, by promoting neoangiogenesis and unregulated T-cell recruitment [?].

5 Comparative Analysis of Identified Associations

5.1 Variants with the Strongest Evidence

The 17q12-21.2 locus (*GSDMB/ORMDL3*, rs7216389) provides the most robust, highly replicated evidence for early-childhood viral-induced exacerbations and hospitalizations. Pharmacogenomic interactions at *ADRB2* (rs1042713) are also strongly validated for LABA-treated pediatric cohorts.

5.2 Variants with Inconsistent Findings

Associations for *TNFA* (rs1800629) and *NR3C1* (Tth111I) show significant inconsistency, heavily influenced by sample size, environmental confounders, and the necessity for multi-locus haplotypic profiling to identify true causal effects.

5.3 Population-Specific Associations

Ancestral genetic architecture heavily dictates risk. *SEMA3D* (rs993312) exhibits extreme population specificity, operating as a severe risk locus in African Americans but remaining non-significant in European cohorts. *CHI3L1* (rs4950928) demonstrates direction-of-effect reversals between Caucasian and Asian populations.

5.4 Pediatric vs Adult Asthma

Pediatric exacerbations are heavily driven by epithelial barrier loci (*CDHR3*, *GSDMB*) and structural adherence defects (*CTNNA3*). In contrast, adult-onset asthma exacerbations involve chronic metabolic pathways, environmental exposures, and HLA/immune progression mechanisms.

5.5 Severe vs Mild Exacerbations

Many loci, such as *CHI3L1* rs4950928, specifically modulate the risk of catastrophic, hospital-requiring events (severe exacerbations) while demonstrating no statistical association with milder out-patient flares requiring only oral corticosteroids.

6 Biological Interpretation of Findings

6.1 Type 2 Inflammation Pathway (*IL4*, *IL4RA*, *IL10*)

Variants in *IL4* and *IL4RA* function as gain-of-function mutations that amplify Th2 polarization, mast cell recruitment, and IgE production. *IL10* promoter deficiencies remove the homeostatic brake on these pathways, predisposing to refractory eosinophilic exacerbations.

6.2 Airway Remodeling Pathway (*TGFB1*, *CTNNA3*, *SEMA3D*)

TGFB1 overexpression drives subepithelial fibrosis, while *CTNNA3* and *SEMA3D* defects impair epithelial adherens junctions and dysregulate angiogenesis. Together, these cause structural airway stiffening and rapid mechanical bronchoconstriction.

6.3 β 2-Adrenergic Signaling (*ADRB2*)

Mutations altering receptor downregulation and tachyphylaxis (*ADRB2*) render airway smooth muscle unresponsive to rescue bronchodilators during active inflammation.

6.4 Corticosteroid Response (*NR3C1*)

Altered glucocorticoid receptor kinetics modify the clinical efficacy of inhaled corticosteroids, promoting localized pro-inflammatory cytokine expression (e.g., IL-15) even under high-dose therapy.

6.5 Metabolic and Adipokine Pathways (*PPARG*, *ADIPOQ*)

Genetic modifications in *PPARG* and *ADIPOQ* facilitate low-grade systemic meta-inflammation, contributing to corticosteroid resistance and exacerbation-prone phenotypes, particularly in obese patients.

6.6 Epithelial Barrier and Innate Immunity (*GSDMB*, *CHI3L1*)

Impaired antiviral responses (e.g., rhinovirus C) and upregulated pyroptosis at the epithelial surface represent primary triggers for severe pediatric asthmatic crises.

7 Clinical Implications

7.1 Precision Medicine in Asthma

Pharmacogenomic stratification using *ADRB2* variants allows for genotype-guided medication selection, successfully reducing exacerbations in trials. Similarly, *IL4RA* typing can predict the efficacy of anti-IL-4R α biologics like dupilumab.

7.2 Genetic Risk Stratification and Biomarkers

While single SNPs lack the positive predictive value for routine clinical screening, integrating robust markers (*GSDMB*, *CDHR3*) into Polygenic Risk Scores (PRS) can proactively identify high-risk pediatric patients requiring aggressive early intervention.

8 Strengths and Limitations

8.1 Strengths

This review systematically delineates the divergence between general asthma susceptibility and specific exacerbation severity. Furthermore, we categorize associations by functional biological pathways rather than isolating individual genes.

8.2 Limitations of Included Studies

Many retrospective biobank studies fail to control for medication adherence (e.g., ICS use), environmental tobacco smoke, or local allergen exposure, introducing significant confounding.

8.3 Limitations of the Review

Language restrictions and the predominant overrepresentation of European-ancestry populations in the original source data limit universal generalizability.

9 Future Research Directions

Future studies must integrate large, multi-ancestry cohorts to account for linkage disequilibrium disparities. Incorporating Whole-Genome Sequencing (WGS), multi-omics approaches (eQTLs, DNA methylation), and rigorous Polygenic Risk Scores (PRS) are essential to map the full epistatic landscape of asthma exacerbations.

10 Conclusions

The genetic architecture of asthma exacerbations is highly distinct, polygenic, and driven by severe structural and immunological pathways. Variants such as *GSDMB* rs7216389, *IL4RA* rs1801275, and *TGFB1* rs1800469 robustly modify tissue vulnerability to acute respiratory flares. Integrating these genomic profiles into clinical practice holds immense potential for realizing personalized, precision-guided asthma management.

Supplementary Materials

Quality of Evidence and SNP Ranking

We conducted an evaluation using the GRADE framework principles to rank the selected SNPs based on evidence level, cross-population consistency, and current clinical utility for predicting asthma exacerbations.

Table 2: SNP Ranking by Evidence and Clinical Utility

SNP	Gene	Evidence Level	Consistency	Clinical Utility
rs7216389	<i>GSDMB</i>	High	High	High
rs6967330	<i>CDHR3</i>	High	High	High
rs1042713	<i>ADRB2</i>	Moderate	Moderate	Moderate (Pharmacogenomic)
rs1801275	<i>IL4RA</i>	Moderate	Moderate	Moderate (Biologics)
rs1800469	<i>TGFB1</i>	Moderate	Low	Low
rs993312	<i>SEMA3D</i>	Moderate	Low (Ancestry-specific)	Low
rs1800896	<i>IL10</i>	Low	Low	Low
rs7915695	<i>CTNNA3</i>	Low	Low	Low

Biological Model of Asthma Exacerbation Susceptibility

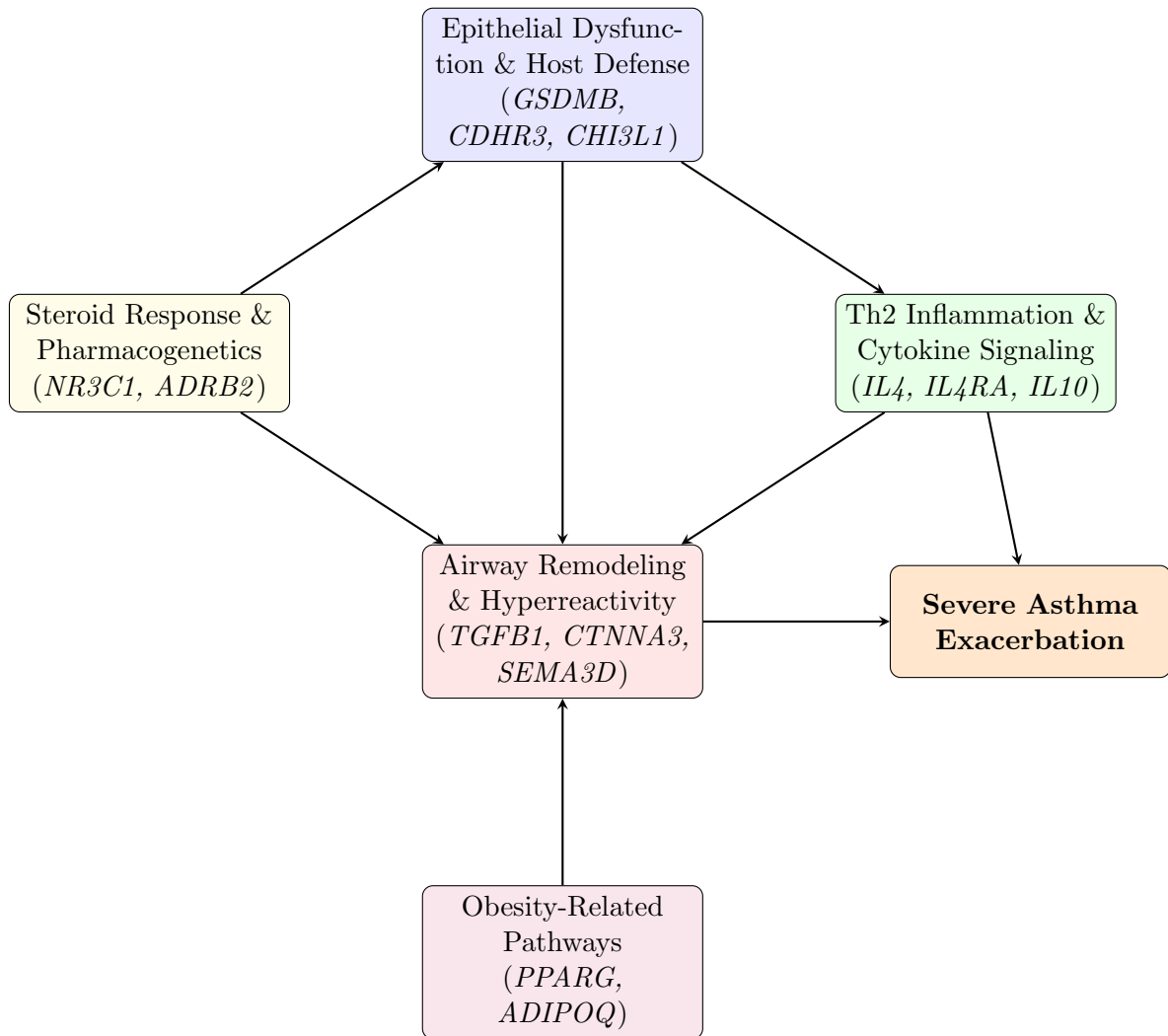


Figure 4: Genetic Architecture of Asthma Exacerbation Susceptibility

Supplementary Tables

Supplementary Table S1: Full Search Strategy

	Database	Search Query
PubMed		(Asthma[MeSH] OR asthma[Title/Abstract]) AND (Exacerbation*[Title/Abstract] OR Severe[Title/Abstract] OR Hospitalization[Title/Abstract]) AND (Polymorphism[Title/Abstract] OR SNP[Title/Abstract] OR Variant[Title/Abstract] OR GWAS[Title/Abstract]) AND (ADRB2 OR IL4 OR IL4RA OR TNFA OR NR3C1 OR PPARG OR TGFB1 OR IL10 OR GSDMB OR CTNNA3 OR ADIPOQ OR SEMA3D)
Embase		('asthma'/exp OR asthma:ti,ab) AND ('disease exacerbation'/exp OR exacerbation:ti,ab OR severe:ti,ab OR hospitalization:ti,ab) AND ('single nucleotide polymorphism'/exp OR SNP:ti,ab OR variant:ti,ab OR GWAS:ti,ab)
GWAS Catalog		(Asthma) AND (Exacerbation)

Supplementary Table S2: Characteristics of Included Studies

Study / Cohort	Design	Population	Genes	Main Finding
CAMP & CARE [?]	GWAS	Pediatric (North America)	<i>CTNNA3</i> , <i>SEMA3D</i>	rs7915695 significantly associated with high exacerbation rates.
COPSAC [?]	Cohort	Pediatric Birth Cohort (Denmark)	<i>GSDMB</i> , <i>CDHR3</i>	<i>GSDMB</i> rs7216389 increased hazard of hospitalizations.
PiCA Consortium [?]	Meta-analysis	Multi-ancestry Pediatric	<i>GSDMB</i>	rs7216389 associated with hospitalizations despite ICS.
UK Biobank [?]	GWAS	Adult (UK)	<i>ADRB2</i>	rs1042713 not associated in adults.
SARP [?]	Cohort	Pediatric/Adult Severe Asthma	<i>GSDMB</i> , <i>ORMDL3</i>	rs2305480 linked to longitudinal exacerbations.
BREATHE, GALA II [?]	Meta-analysis	Pediatric	<i>ADRB2</i>	<i>ADRB2</i> Arg16 increased AE risk with LABA.
BioVU [?]	EHR Cohort	Multi-ancestry (USA)	<i>SEMA3D</i>	<i>SEMA3D</i> risk specific to African American cohort.

Supplementary Table S3: Risk-of-bias assessment (Newcastle-Ottawa Scale summary)

Study Type	Selection (0-4)	Comparability (0-2)	Outcome (0-3)
Prospective Cohorts (e.g. COPSAC)	4	2	3
Clinical Trials (e.g. CAMP)	4	2	3
Retrospective EHR (e.g. BioVU)	3	1	2
Meta-analyses (Overall Quality)	High	High	High

Supplementary Table S4: Summary of all SNP associations

SNP	Gene	Clinical Association Outcome
rs1042713	<i>ADRB2</i>	Exacerbation risk with ICS+LABA in children.
rs1042714	<i>ADRB2</i>	Severe uncontrolled asthma with ICS.
rs2243250	<i>IL4</i>	Fatal/near-fatal exacerbations.
rs1801275	<i>IL4RA</i>	Exacerbations requiring ICU admission.
rs1805011	<i>IL4RA</i>	High IgE / severe asthmatic phenotypes.
rs1800629	<i>TNFA</i>	Severe asthmatic crises, AERD.
rs10052957	<i>NR3C1</i>	Weak/regional association, glucocorticoid resistance if co-inherited.
rs6195	<i>NR3C1</i>	AA genotype = severe uncontrolled asthma.
rs1801282	<i>PPARG</i>	Rare allele is protective.
rs3856806	<i>PPARG</i>	Rare allele is protective.
rs1800469	<i>TGFB1</i>	Asthma severity and airway remodeling.
rs2241712	<i>TGFB1</i>	Exacerbations with high HDM exposure.
rs1800896	<i>IL10</i>	A allele increases hospitalizations; G is protective.
rs1800871	<i>IL10</i>	Part of ATA severe asthma haplotype.
rs1800872	<i>IL10</i>	Part of ATA severe asthma haplotype.
rs4950928	<i>CHI3L1</i>	G allele protects against hospitalizations.
rs12141494	<i>CHI3L1</i>	AA predicts progressive airway remodeling.
rs7216389	<i>GSDMB</i>	High hazard of early childhood exacerbations.
rs2305480	<i>GSDMB</i>	Severe asthma, epistatic with CDHR3.
rs4794820	<i>ORMDL3</i>	Severe refractory asthma.
rs7915695	<i>CTNNA3</i>	High exacerbation rate in children.
11377C ₆ G	<i>ADIPOQ</i>	Obesity, meta-inflammation, steroid resistance.
rs993312	<i>SEMA3D</i>	Severe exacerbations in African Americans.

Supplementary Figure S1: PRISMA flow diagram.

This diagram is presented as Figure ?? in the main text.